

Project Workflow: Week 3

Phase: Environment Setup & Project Initialization

1. Objective

The primary goal of Week 3 was to establish the complete Java development environment and initialize the Ping Pong Game project structure. This included:

- Installing and configuring the Java Development Kit (JDK).
- Setting up the selected Integrated Development Environment (IDE).
- Configuring Java Swing for graphical user interface development.
- Creating the initial project structure and main game window.
- Preparing the foundation for future game development.

2. Detailed Task Breakdown

JDK Installation & IDE Configuration

Installed the required Java Development Kit and configured the development environment. The IDE was prepared for Java development, including project SDK configuration and basic execution testing.

Java Swing Environment Setup

Configured Java Swing and AWT libraries for developing the game's graphical interface. Basic Swing components such as JFrame and JPanel were tested to ensure the GUI environment was working correctly.

Project Structure Creation

Created the initial project structure and organized the Java source files and classes. A modular structure was prepared so that future features such as paddle movement, ball physics, collision detection, and scoring could be added easily.

Main Game Window Development

Created the main PingPongGame class using JFrame. The basic game window was configured with an appropriate title, size, closing operation, and screen position.

3. Deliverables

- **Configured Development Environment:** JDK and IDE successfully installed and configured.
- **Swing Environment:** Java Swing/AWT environment prepared for GUI development.
- **Initial Project Structure:** Basic Java project and class organization created.
- **Main Game Window:** Functional PingPongGame JFrame created.

- **Environment Test:** Basic Java/Swing program successfully executed.

4. Tools & Resources

Development Environment

- Java JDK 11+
- IntelliJ IDEA / VS Code

Libraries & Technologies

- Java Swing
- Java AWT
- JFrame
- JPanel